

- Q.30 Explain with example, each mode of heat transfer.
- Q.31 How does Reynolds number relate to fluid flow and heat transfer?
- Q.32 Define the convective heat transfer coefficient and discuss the factors influencing its value. Provide examples where the convective heat transfer coefficient plays a crucial role.
- Q.33 Explain about Planck's law, and Kirchoff's law.
- Q.34 Write down about Absorptivity and reflectivity.
- Q.35 With diagram, write about open pan evaporator.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the critical role of insulation in maintaining temperature control in a specific industrial process, considering factors like material selection and thickness.
- Q.37 Derive one dimensional steady state heat conduction through a sphere.
- Q.38 Write short note on any three:-
1. Emissivity
 2. Optimum thickness of insulation
 3. Mechanism of Conduction.
 4. Thermal conductivity of materials-solids, liquids, and gases

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Roll No.

3rd Sem / Chem, P&P Subject:- Heat Transfer - I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which heat transfer mechanism occurs through the movement of fluid particles?
- a) Conduction
 - b) Convection
 - c) Radiation
 - d) Both A and C
- Q.2 The composite wall in heat conduction involves:
- a) Multiple materials in layers
 - b) A single homogeneous material
 - c) A mixture of solids and liquids
 - d) A combination of conduction and convection
- Q.3 Newton's law of cooling is associated with which mode of heat transfer?
- a) Conduction
 - b) Convection
 - c) Radiation
 - d) All of the above
- Q.4 Which dimensional group is important in forced convection and relates fluid velocity to characteristic length and fluid properties?
- a) Reynolds number
 - b) Nusselt number
 - c) Prandtl number
 - d) Grasshof number
- Q.5 What does emissivity measure in radiation heat transfer?

- a) Reflectivity b) Transmissivity
c) Absorptivity d) All of the above
- Q.6 The critical thickness of insulation is mainly dependent on:
a) Material conductivity
b) Surface area
c) Temperature difference
d) Insulation color
- Q.7 Which law is fundamental in understanding radiation between surfaces and is expressed as the ratio of the actual emissive power to the maximum possible emissive power?
a) Stefan - Boltzmann Law
b) Kirchhoff's Law
c) Wien's displacement law
d) View factor law
- Q.8 What is the primary difference between steady-state and unsteady-state heat transfer?
a) Temperature remains constant in steady-state
b) Temperature varies continuously in steady-state
c) Unsteady-state involves only convection
d) Unsteady-state is more efficient
- Q.9 Fourier law of heat conduction is best represented by
a) $Q = -kA \frac{dt}{dx}$ b) $Q = kA \frac{dx}{dt}$
c) $Q = -kA$ d) $Q = k \frac{dt}{dx}$
- Q.10 Reynolds number is _____
a) Dimensionless number
b) Dimensional number
c) Both A & B
d) None of above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is black body?
Q.12 What is the Prandtl number?
Q.13 Define emissivity?
Q.14 Write name of any one heat exchange?
Q.15 What is conductor.
Q.16 What is radiation heat transfer?
Q.17 Name a common insulating material.
Q.18 Give an example of natural convection.
Q.19 What is Wein's displacement law?
Q.20 Define thermal conductivity.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain about boiling, condensation and evaporation.
Q.22 Write about optimum thickness of insulation.
Q.23 Discuss one dimensionless steady state heat conduction through composite wall?
Q.24 Describe about physical significance of dimensionless number.
Q.25 Differentiate between free and forced convection.
Q.26 What are the key mechanisms of heat transfer.
Q.27 What is dimensional analysis used for in heat transfer.
Q.28 What does the effect of temperature on thermal conductivity.
Q.29 Briefly explain the purpose of a radiation shield.

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